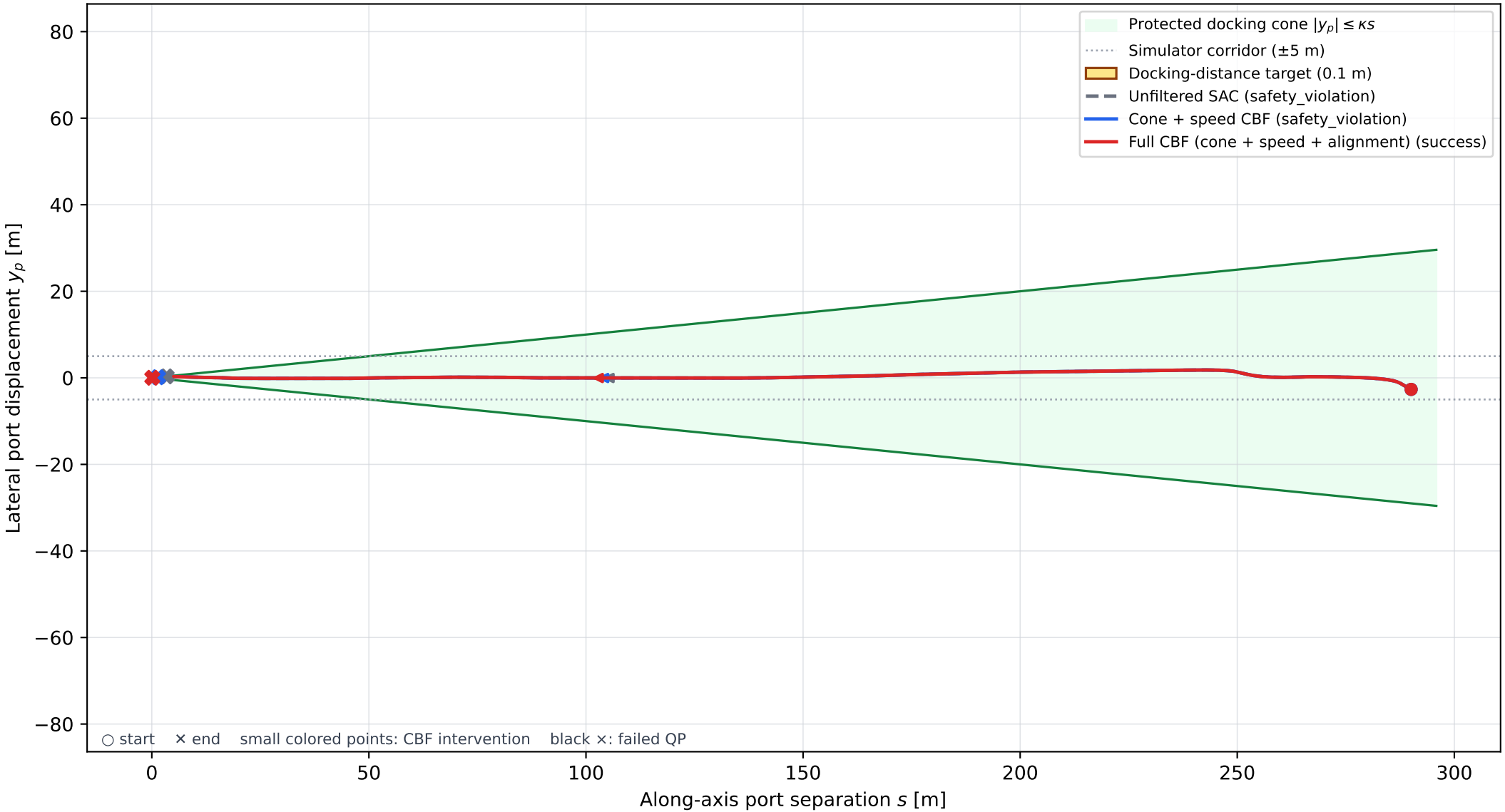
Final-approach spatial trajectory — evaluation seed 120
Protected docking cone $|y_p| \le \kappa s$
Simulator corridor (±5 m)
Docking-distance target (0.1 m)
Unfiltered SAC (safety_violation)
Cone + speed CBF (safety_violation)
Full CBF (cone + speed + alignment) (success)
Lateral port displacement y_p [m]
Along-axis port separation s [m]
80
60
40
20
0
−20
−40
−60
−80
0
50
100
150
200
250
300
○ start × end small colored points: CBF intervention black ×: failed QP